

## SELECT with various clauses – BETWEEN, IN, Aggregate Function

In& between: List the type of the courses “Statistics” and “Programming”

Aggregate: List the maximum, minimum, average mark for each course.

```
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 29
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database university;
Query OK, 1 row affected (0.08 sec)

mysql> use university;
Database changed
mysql> create table course(courseid int primary key,coursename varchar(30),coursetype varchar(20));
Query OK, 0 rows affected (0.16 sec)

mysql> create table marks(regno int,courseid int,marks int);
Query OK, 0 rows affected (0.09 sec)

mysql> insert into course values(101,'statistics','theory'),(102,'programming','practical'),(103,'DBMS','theory'),(104,'java','practical');
Query OK, 4 rows affected (0.06 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> insert into marks values(191711341,101,85),(191711342,101,90),(191711343,101,75),(191711341,102,88),(191711342,102,95),(191711343,102,80),(191711341,103,78),
(191711342,103,82),(191711343,103,70);
Query OK, 9 rows affected (0.07 sec)
Records: 9  Duplicates: 0  Warnings: 0

mysql> select*from course;
+-----+-----+-----+
| courseid | coursename | coursetype |
+-----+-----+-----+
| 101 | statistics | theory |
| 102 | programming | practical |
| 103 | DBMS | theory |
| 104 | java | practical |
+-----+-----+-----+
4 rows in set (0.05 sec)
```

```
104 | java | practical |
+-----+-----+-----+
4 rows in set (0.05 sec)

mysql> select*from marks;
+-----+-----+-----+
| regno | courseid | marks |
+-----+-----+-----+
| 191711341 | 101 | 85 |
| 191711342 | 101 | 90 |
| 191711343 | 101 | 75 |
| 191711341 | 102 | 88 |
| 191711342 | 102 | 95 |
| 191711343 | 102 | 80 |
| 191711341 | 103 | 78 |
| 191711342 | 103 | 82 |
| 191711343 | 103 | 70 |
+-----+-----+-----+
9 rows in set (0.00 sec)

mysql> select coursename,coursetype from course where coursename in ('statistics','programming');
+-----+-----+
| coursename | coursetype |
+-----+-----+
| statistics | theory |
| programming | practical |
+-----+-----+
2 rows in set (0.05 sec)

mysql> select*from marks where marks between 80 and 90;
+-----+-----+-----+
| regno | courseid | marks |
+-----+-----+-----+
| 191711341 | 101 | 85 |
| 191711342 | 101 | 90 |
| 191711341 | 102 | 88 |
| 191711343 | 102 | 80 |
| 191711342 | 103 | 82 |
+-----+-----+-----+
5 rows in set (0.05 sec)
```

2 rows in set (0.05 sec)

mysql> select \* from marks where marks between 80 and 90;

| regno     | courseid | marks |
|-----------|----------|-------|
| 191711341 | 101      | 85    |
| 191711342 | 101      | 90    |
| 191711341 | 102      | 88    |
| 191711343 | 102      | 80    |
| 191711342 | 103      | 82    |

5 rows in set (0.05 sec)

mysql> select courseid, max(marks) as maximum, min(marks) as minimum, avg(marks) as average from marks group by courseid;

| courseid | maximum | minimum | average |
|----------|---------|---------|---------|
| 101      | 90      | 75      | 83.3333 |
| 102      | 95      | 80      | 87.6667 |
| 103      | 82      | 70      | 76.6667 |

3 rows in set (0.06 sec)